Functions

Part A: Notation

Q 1 For the function $f(x)=4x-3$, find:

(i) the value of $f(3a-1)$. 1

(ii) the value of x at which $f(x)=0$. 1

13(e)(i)

$f(x)=4x-3$

$f(3a-1)=4\times(3a-1)-3$

$=12a-4-2$

$=12a-6$

13(e)(ii)

$f(x)=4x-3$

$0=4x-3$

$4x=3$

$x=\frac{3}{4}$

Q 2 If $f(x)=5x^2-2x+2$ and $g(x)=3x+2$ then for what values of x is $f(x)=g(x)$? 2

$5x^2-2x+2=3x+2$

$5x^2-5x=0$ ✓

$5x(x-1)=0$

$x=0$ or $x=1$ ✓

Q 3 If $f(x)=2x^2+5x+1$ calculate $f(-3)$. 2

$f(-3)=2\times-3^2+5\times-3+1$ ✓

$=18-15+1$

$=4$ ✓

Q 4 If $g(x)=x^2+1$

i). Evaluate $g(-3)$. 1

ii). For what values of x is $g(x)=2$? 2

$g(x)=x^2+1$

i) $g(-3)=(-3)^2+1$

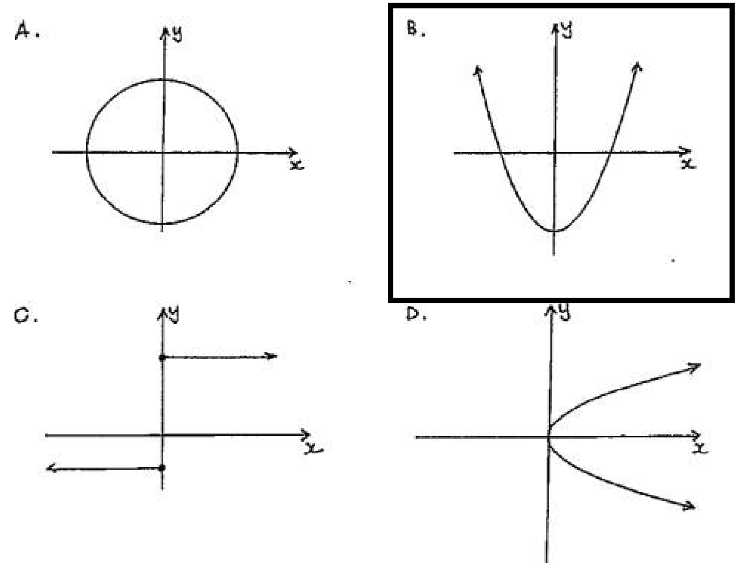
$=10$

ii $x^2+1=2$

$x^2=1$

$x=\pm1$

Q 5 Which of the following graphs represents a function?



Q 6

If $f(x) = x^2 + 1$:

(i) Evaluate $f(-5)$. **1**

(ii) For what value(s) of x is $f(x) = 5$? **2**

(b) $f(x) = x^2 + 1$

(i) $f(-5) = (-5)^2 + 1$
$= 25 + 1$
$= 26$

(ii) $5 = x^2 + 1$
$x^2 = 4$
$x = \pm 2$

Q 7

Consider $f(x) = \frac{6}{x}$ and $g(x) = 2x + 4$. What are the values for x for which $f(x) = g(x)$?

(A) $x = -1$ or $x = 3$

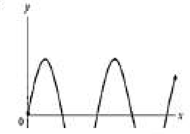
(B) $x = -1$ or $x = -3$

(C) $x = 1$ or $x = 3$

(D) $x = 1$ or $x = -3$

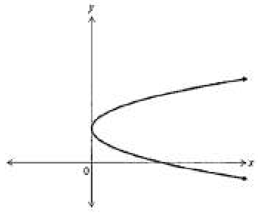
Q 8

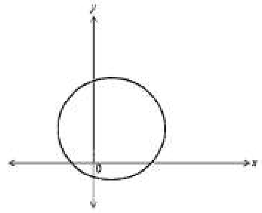
Which of the following graphs represents a function?

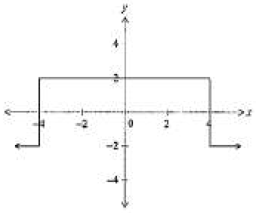
(A)


A

it passes the vertical line test

(B)


(C)


(D)


Q 9

Given that $f(x) = 2x + 4$,

(i) Find a if $f(a) = -2$ **2**

(ii) Find $\frac{f(2x)}{4}$ **2**

Given that $f(x) = 2x + 4$,

(i)
$2a + 4 = -2$
$a = -3$

(ii)
$\frac{f(2x)}{4}$
$= \frac{2(2x) + 4}{4}$
$= \frac{4(x+1)}{4}$
$= x + 1$

Q 1

Which type of function is $f(x) = 5x^2 - x$?

(A) Odd
(B) Even
(C) Neither odd or even
(D) Zero

$$f(-x) = 5(-x)^2 - (-x)$$
$$= 5x^2 + x$$
$$\neq f(x) \text{ or } -f(x)$$

Therefore the function is neither odd or even.

1 Mark: C

Q 2

Determine if the function $f(x) = \dfrac{1}{(x-1)(x+1)}$ is an odd or even function, or neither. **2**

Justify your answer.

$$f(x) = \frac{1}{(x-1)(x+1)} = \frac{1}{x^2-1}$$
$$f(-x) = \frac{1}{(-x-1)(-x+1)} = \frac{-1}{(x+1)(-x+1)}$$
$$= \frac{-1}{1-x^2} = \frac{-1}{-(x^2-1)} = \frac{1}{x^2-1}$$

$\therefore$ even.

Q 3

Consider the function $y = \sqrt{16 - x^2}$

Show that it is an even function 1

$$f(-x) = \sqrt{16-(-x)^2}$$
$$= 16 - x^2$$
$$= f(x)$$

Hence it is an even function

Q 4

Explain why the function $f(x) = 3x^2 - 2x + 6$ is neither odd nor even. **1**

$$f(x) = 3x^2 - 2x + 6$$
$$f(-x) = 3(-x)^2 - 2(-x) + 6$$
$$f(-x) = 3x^2 + 2x + 6$$
$$f(-x) \neq f(x) \neq -f(x)$$

$\therefore$ Neither odd of Even.

Q 5 Is the function $f(x) = 8x^3 - 7x$ even, odd or neither? **2**
(Justify you answer)

$f(x) = 8x^3 - 7x$

$f(-x) = 8(-x)^3 - 7(-x)$

$= -8x^3 + 7x$

$= -(8x^3 - 7x)$

$= -f(x)$

As $f(-x) = -f(x)$, the function is odd.

Q 6 Explain why $f(x) = 3^x$ is neither an odd function nor an even function. **1**

$f(x) = 3^x$

$f(-x) = 3^{-x}$

$= \frac{1}{3^x}$

$\neq f(x)$ or $-f(x)$

$\therefore$ the function is neither.

or draw graph and explain that it is not symmetrical about y axis and has no point symmetry

Q 7 Show that the function $f(x) = \frac{1 - x^2}{x}$ is an odd function. **2**

i) $f(x) = \frac{1 - x^2}{x}$

$f(-x) = \frac{1 - (-x)^2}{(-x)}$

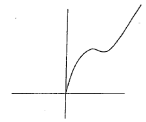
$= \frac{1 - x^2}{-x} = \frac{-(1 - x^2)}{x}$

$-f(x) = \frac{-(1 - x^2)}{x}$

$\therefore f(-x) = -f(x)$

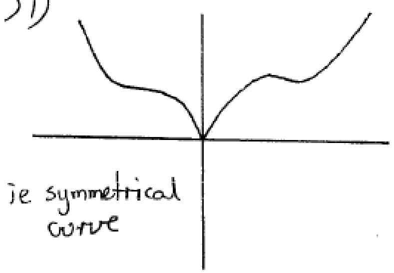
ODD

Q 8 Copy this figure to the right onto your exam page.

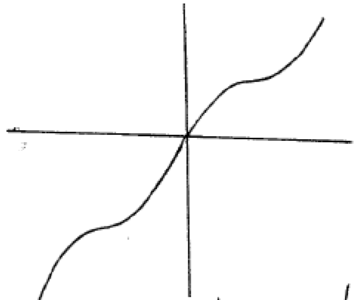


i. Using a dotted line, complete f(x) if f(x) is an even function. 1 m
ii. Using a solid line, complete f(x) if f(x) is a n odd function. 1 m

i) ii)



ie symmetrical curve



ie has point symmetry.

Q 9 Is the function $f(x) = x^4 + 2x$ odd, even or neither? Justify your answer. 3

(i) $f(x) = x^4 + 2x$

$f(-x) = (-x)^4 + 2(-x)$

$= x^4 - 2x$

$f(-x) \neq f(x)$ 3

and $f(-x) \neq -f(x)$

$\therefore f(x)$ is neither even nor odd function

Q10 Show algebraically that $f(x) = \dfrac{x^3}{x^4 - x^2}$ is an **odd** function. 2

Q4/ (a) $f(x) = \dfrac{x^3}{x^4 - x^2}$ $f(-x) = \dfrac{(-x)^3}{(-x)^4 - (-x)^2}$

$= -\dfrac{(x)^3}{x^4 - x^2}$

$\therefore f(x) = -f(-x)$

$\therefore$ odd

Q11 Is the function $f(x) = \dfrac{x}{x^2 - 1}$ odd, even or neither? Justify your answer. 3

(i) $f(-x) = \dfrac{-x}{(-x)^2 - 1}$

$= \dfrac{-x}{x^2 - 1}$

$= -\left(\dfrac{x}{x^2 - 1}\right)$ 3

$= -f(x)$

$\therefore$ $f(x)$ is an odd function.

Q12 Which of the following is true for the function

$f(x) = \frac{3}{x^2} - 2$?

A. Even function

B. Odd function

C. Neither odd nor even function

D. Zero function

Since $f(x) = f(-x)$

$\therefore$ even function A

Q13

Determine whether the function $f(x)=\dfrac{1}{x^2-4}$ is odd, even or neither odd nor even. WORKING MUST BE SHOWN. 1

$f(x) = \dfrac{1}{x^2-4}$

EVEN

$f(-x) = \dfrac{1}{(-x)^2-4} = \dfrac{1}{x^2-4}$

$\therefore$ even since $f(x) = f(-x)$

Q14

Which of the following is true for the function $f(x)=\dfrac{3}{x^2}-2$?

A. Even function

B. Odd function

C. Neither odd nor even function

D. Zero function.

A

$$f(x)=\frac{3}{x^2}-2$$

$$f(a)=\frac{3}{a^2}-2$$

$$f(-a)=\frac{3}{(-a)^2}-2$$

$$=\frac{3}{a^2}-2$$

$f(-a)=f(a)$ $\quad\therefore$ An even function

Q15

Determine if the function below is odd, even or neither: 2

$$f(x)=\frac{3x}{2+x^2}$$

(c) $f(x) = \dfrac{3x}{2+x^2}$

$-f(x) = \dfrac{-3x}{2+x^2}$

$f(-x) = \dfrac{3(-x)}{2+(-x)^2}$

$= \dfrac{-3x}{2+x^2}$

ie $f(-x) = -f(x)$ $\quad\therefore$ odd

Q16

Given $f(x) = \dfrac{x^3-4x}{x^2+4}$, 2

is $f(x)$ an even function, odd function, or neither? Justify your answer.

$f(-x) = \dfrac{(-x)^3-4(-x)}{(-x)^2+4}$

$= \dfrac{-x^3+4x}{x^2+4}$

$= -\dfrac{x^3-4x}{x^2+4}$

① mark

$f(-x) = -f(x)$

$\therefore$ odd

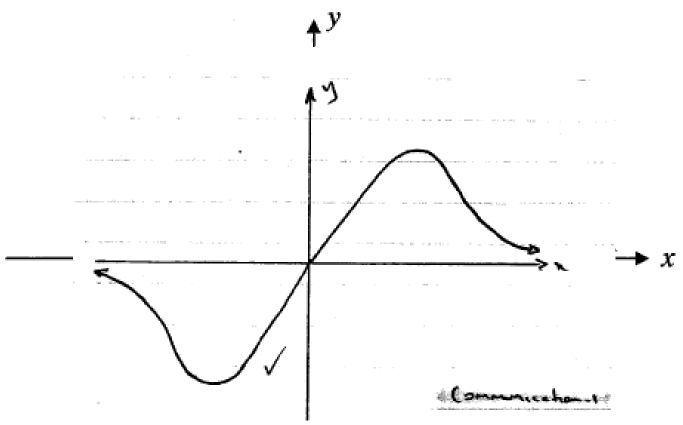
① Correct answer according to finding

Q17 If $f(x)=2x^2-x$ is this an odd function, even function or neither? **1**

$$
\begin{aligned}
f(x) &= 2x^2 - x \\
f(-x) &= 2(-x)^2 - (-x) \\
&= 2x^2 + x \\
&\neq \pm f(x)
\end{aligned}
$$

$\therefore$ *neither even or odd*

Q18 The diagram below shows part of the graph of $y=f(x)$.
You are told that $f(x)$ is an odd function.



Copy the diagram and complete the graph of the function.

Q 1

The function $y = f(x)$ is defined as follows:

$$f(x) = \begin{cases} -3 & \text{for } x \le -1 \\ x+1 & \text{for } -1 < x < 3 \\ x^2+1 & \text{for } x \ge 3 \end{cases}$$

(i) Evaluate $f(-1) + f(3)$. 1

(ii) Write an expression for $f(a^2+3)$. 1

11(f) (i)	$f(-1)+f(3) = -3+(3^2+1)$ $=7$
11(f) (ii)	Since $(a^2+3) \ge 3$ $f(a^2+3) = (a^2+3)^2+1$ $= a^4+6a^2+9+1$ $= a^4+6a^2+10$

Q 2

The function $y = f(x)$ is defined as follows:

$$f(x) = \begin{cases} x-1 & \text{for } x \le -2 \\ -1 & \text{for } -2 < x < 1 \\ x+1 & \text{for } x \ge 1 \end{cases}$$

(i) Evaluate $f(-2) + f(1)$. 1

(ii) Write an expression for $f(a^2+1)$. 1

12(b) (i)	$f(-2)+f(1) = (-2-1)+(1+1)$ $=-1$
12(b) (ii)	Since $(a^2+1) \ge 1$ $f(a^2+1) = a^2+1+1$ $= a^2+2$

Q 3

A function is defined as

$$f(x) = \begin{cases} x-5 & x \ge 0 \\ -2 & -3 < x < 0 \\ 2+x & x \le -3 \end{cases}$$

Find

(i) $f(-1) + f(-5)$ 1

(ii) $f\left(a^2\right)$ 1

i. $f(-1) + f(-5) = [-2] + [2-5]$
$= -2 + -3$
$= -5$

ii. $f(a^2) = a^2 - 5$

Q 4

A function is defined as: $f(x) = \begin{cases} -x^2 & \text{when } x \le 0 \\ 5x-4 & \text{when } x > 0 \end{cases}$

Evaluate f(-2) + f(3) - f(0). 2

$$f(x) = \begin{cases} -x^2 & x \le 0 \\ 5x-4 & x > 0 \end{cases}$$

$f(-2) = -(-2)^2 = -4$

$f(3) = 5(3) - 4 = 8$

$f(0) = -(0)^2 = 0$

$-4 + 8 - 0 = 4$

Q 5

If $f(x) = \begin{cases} 0 & if \quad x \le 0 \\ -1 & if \quad 0 < x < 2 \\ x & if \quad x \ge 2 \end{cases}$

Evaluate $f(-1) + f(1) + f(5)$ 2

$(0) + (-1) + (5) = 4$

Q 6

Consider $f(x)=\begin{cases}-3 & \text{if } x\le -1\\ 2x-3 & \text{if } x>-1\end{cases}$. Evaluate $f(-1)+f(1)$. 1

$f(x)=\begin{cases}-3 & x\le -1\\ 2x-3 & x>-1\end{cases}$

$f(-1) = -3$ 1

$f(1) = 2(1)-3 = -1$

$\therefore f(-1)+f(1) = -3+(-1)$

Q 7

A function is defined by the rule $g(x)=\begin{cases}x+1, & \text{if } x\ge 1\\ -1, & \text{if } -2<x<1\\ 1-x, & \text{if } x\le -2\end{cases}$ 4

Find if they exist,

i) $g(1)$ iii) $g(0)$

ii) $g(-1)$ iv) $g(2)+g(-2)$

b) i)
$g(1) = 1+1$
$= 2$

ii)
$g(-1) = -1$

iii)
$g(0) = -1$

iv)
$g(2)+g(-2) = (2+1)+(1-(-2))$
$= 3+3$
$= 6$

Q 8

A piecemeal function is defined as

$f(x)=\begin{cases}-x^2, & x<0\\ 5x-4, & x\ge 0\end{cases}$

Evaluate $f(-2)+f(0)-f(1)$ 2

a) $f(-2)+f(0)-f(1)$
$= -4 + -4 - 1$
$= -9$

Q 9

A function is defined as:

$f(x)=\begin{cases}4-x & \text{for } x<-2\\ 5 & \text{for } -2\le x<0\\ x^3+1 & \text{for } x\ge 0\end{cases}$

Calculate the value of:

(i) $f(-10)$ 1

(ii) $f(-2)$ 1

(iii) $f(m^2)$ 1

i) $f(-10) = 4-(-10)$
$= 14$ ✓

ii) $f(-2) = 5$ ✓

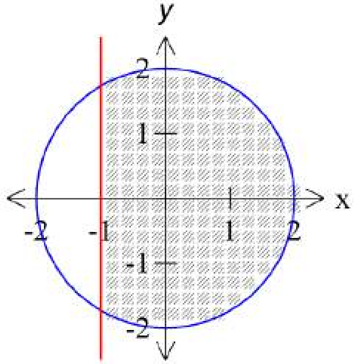
iii) $f(m^2) = (m^2)^3+1$
$= m^6+1$ ✓

Functions
Part D: graphs

2unit

Selective

Q 1 Draw a sketch showing the region on the number plane where the inequalities $x^2+y^2\le 4$ and $x\ge -1$ hold simultaneously. **3**



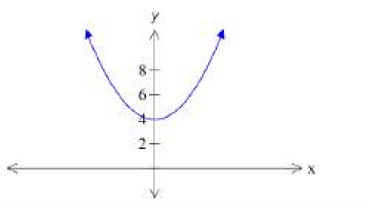
Q 2 Make neat sketches of the following equations on separate sets of axes. Mark clearly the essential features of each graph.

(i) $y=x^2+4$ **1**

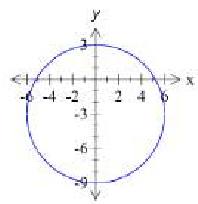
(ii) $x^2+(y+3)^2=36$ **1**

(iii) $y=2^{-x}$ **1**

11(g) (i)

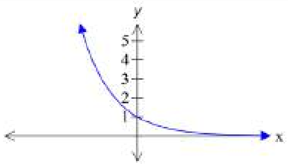


11(g) (ii)



Circle centre $(0,-3)$ and radius 6

11(g) (iii)



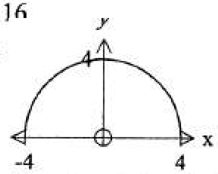
Q 3 What is the domain and range of the function $f(x)=\sqrt{16-x^2}$?

(A) Domain: $-4\le x\le 4$, Range: $0\le y\le 4$

(B) Domain: $-4\le x\le 4$, Range: $-4\le y\le 4$

(C) Domain: $0\le x\le 16$, Range: $-16\le y\le 16$

(D) Domain: $0\le x\le 16$, Range: $0\le y\le 16$



1 Mark: A

Domain: $-4\le x\le 4$, Range: $0\le y\le 4$

Q 4 Make neat sketches of the following equations on separate sets of axes. Mark clearly the essential features of each graph.

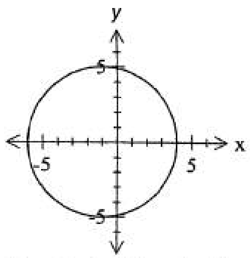
(i) $(x+1)^2+y^2=25$ **1**

(ii) $y=4-x^2$ **1**

(iii) $xy=2$ **1**

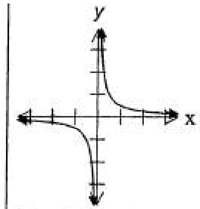
(iv) $y=-2^{-x}$ **1**

13(a) (i)



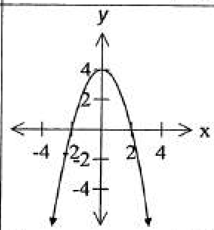
Circle with centre $(-1,0)$ and radius 5

13(a) (iii)



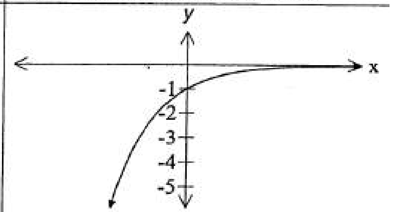
Hyperbola showing the asymptotes

13(a) (ii)



Concave down parabola with vertex (0,4)

13(a) (iv)



Exponential curve passing through $(0,-1)$

Q 5 What is the domain and range of the function $y=\sqrt{4-x}$?

(A) $x\le 4;\ y\ge 0$

(B) $0\le x\le 4;\ y\ge 0$

(C) $x\ge 0;\ y\ge 0$

(D) All real x, all real y.

$y=\sqrt{4-x}$

$y^2=4-x$

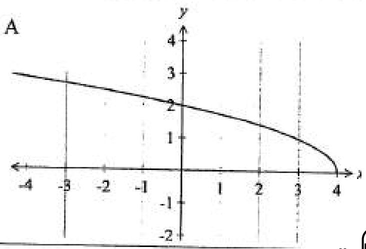
Top half of a parabola.

Passes through (0,2) (4,0)

Domain: $x\le 4$

Range: $y\ge 0$

1 Mark: A

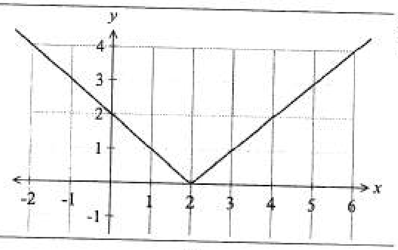


Q 6 Make neat sketches of the following equations on separate sets of axes. Mark clearly the essential features of each graph.

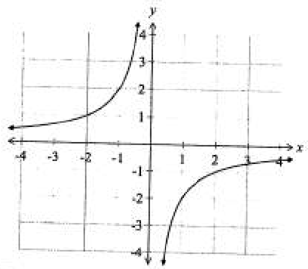
(i) $y=|x-2|$ (ii) $y=1+x^2$ **2**

(iii) $xy=-2$ (iv) $y=-3^x$ **2**

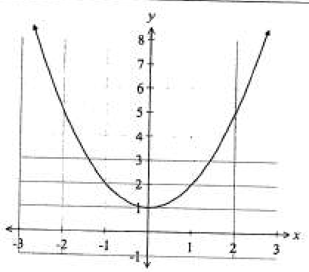
13(d) (i)



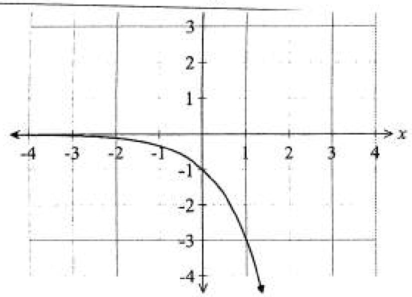
3(d) iii)



13(d) (ii)



(d) /)

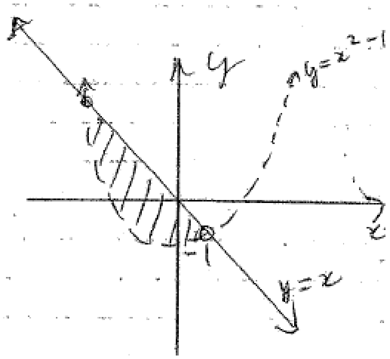


Q 7 Find the largest possible domain for the function $y=\sqrt{2x-3}$ **1**

$2x-3\ge 0$

$x\ge\frac{3}{2}$ ✓

Q 8 On a number plane sketch the region where $y>x^2-1$ and $y\le x$ **2**



Q 9 For what values of x is the function $f(x)=\dfrac{x+2}{2x-1}$ undefined ? **1**

$f(x)=\dfrac{x+2}{2x-1}$

undefined for: $2x-1=0$

$x=\frac{1}{2}$

Q10 What is the natural domain of the function $g(x)=\sqrt{4-2x}$? 2

$g(x)=\sqrt{4-2x}$

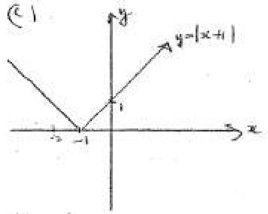
$4-2x \geq 0$

$2x \leq 4$

$x \leq 2$

domain is $x \leq 2$

Q11 Graph the function $y=|x+1|$ and state its domain and range. 4



Domain : $x \in R$

Range : $y \geq 0$

Q12 State the domain and range for each function

i) $y=\sqrt{4-x^2}$ 2

ii) $y=\dfrac{1}{(x^2-1)}$ 2

i) $D: x \leq 2 \;\; or \; -2 \leq x \leq 2$ 1

$R: y \geq 0$ 1

ii) D: all x except $x=\mp 1$ 1

R: $y>0, y<-1$ 1

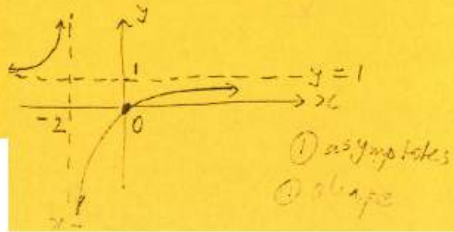
Q13 i) Show that $\dfrac{x}{x+2}=1-\dfrac{2}{x+2}$ 1

ii) Hence or otherwise, sketch the following graph showing all important features $y=\dfrac{x}{x+2}$ 2

a) i) $\dfrac{x}{x+2}=1-\dfrac{2}{x+2}$ ii) $y=\dfrac{x}{x+2}=1-\dfrac{2}{x+2}$

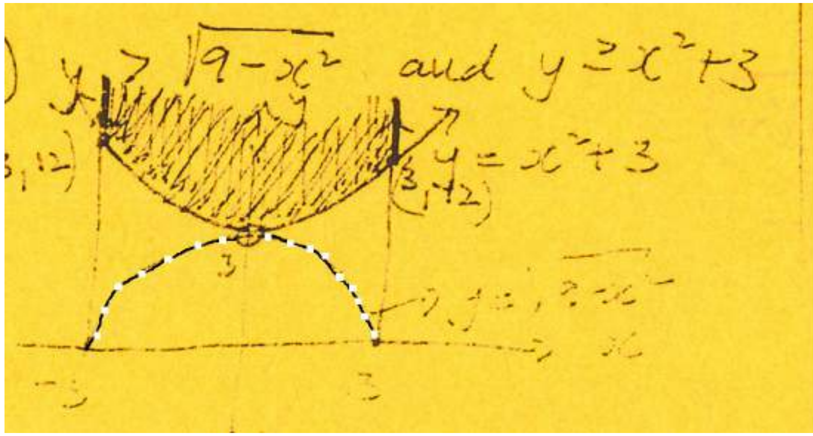
$RHS = 1-\dfrac{2}{x+2} = \dfrac{x+2-2}{x+2}$

$= \dfrac{x}{x+2} = LHS$



Q14 On a number plane, shade the region satisfying both the inequalities 3

$$y>\sqrt{9-x^2} \quad and \quad y\geq x^2+3$$



Q15

Consider the function $y = \sqrt{16 - x^2}$

State the domain. 1

$-4 \leq x \leq 4$ ✓

Q16

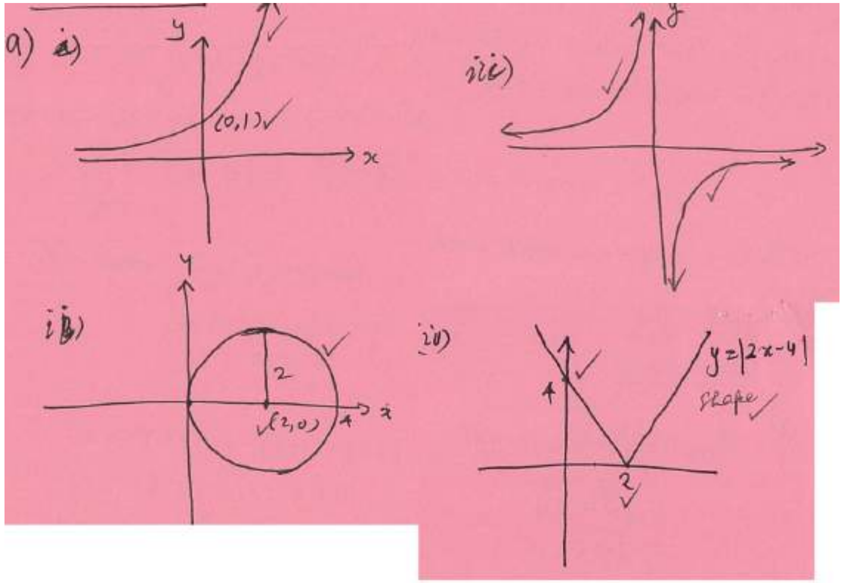
Sketch each of the following showing all important features: 9

i) $y = 3^x$

ii) $(x-2)^2 + y^2 = 4$

iii) $y = \dfrac{-1}{x}$

iv) $y = |2x - 4|$



Q17

The domain and range for the function $y = \sqrt{7 - x}$ is

(A) $x \leq 7$; $y \geq 0$

(B) $0 \leq x \leq 7$; $y \geq 0$

(C) $x \geq 0$; $y \geq 0$

(D) All real x, All real y

Q18

The equation $x^2 + y^2 - 6x + 4y + 6 = 0$ represents a circle. The coordinates of the centre of the circle are:

(A) (3, –2)

(B) (–3, 2)

(C) (–3, –2)

(D) (3, 2)

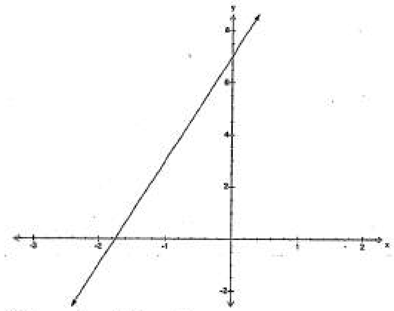
Q19

Sketch the graphs of the following, stating the domain and range of each.

i) $4x - y + 7 = 0$ 3

ii) $x^2 + y^2 = 36$ 3

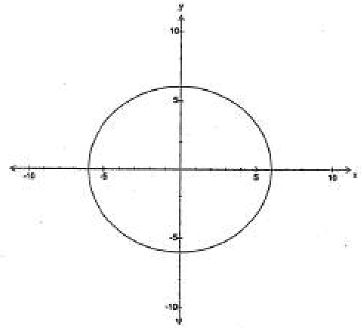
i)



Domain: All real x

Range: All real y

ii)

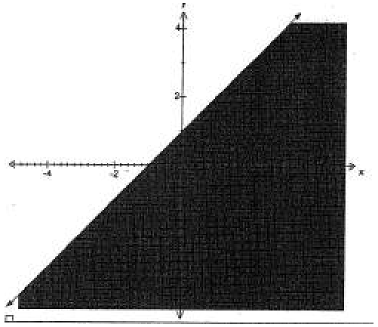


Domain: $-6 \leq x \leq 6$

Range: $-6 \leq y \leq 6$

Q20 Shade the region of the number plane where the following holds: **2**

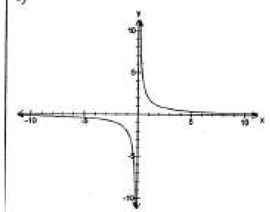
$y \le x + 1$



Q21 Sketch the graphs of the following, stating the domain and range of each.

i) $y = \dfrac{2}{x}$ **2** ii) $x^2 + y^2 = 25$ **2** iii) $3(x+2) - y = 0$ **2**

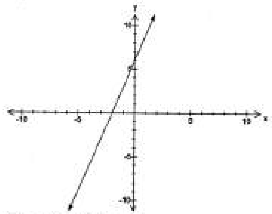
i)



Domain: All real x ; $x \ne 0$

Range: All real y ; $y \ne 0$

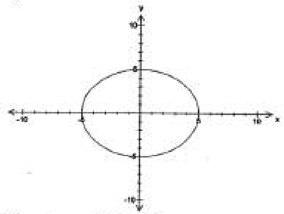
iii)



Domain: All real x

Range: All real y

ii)

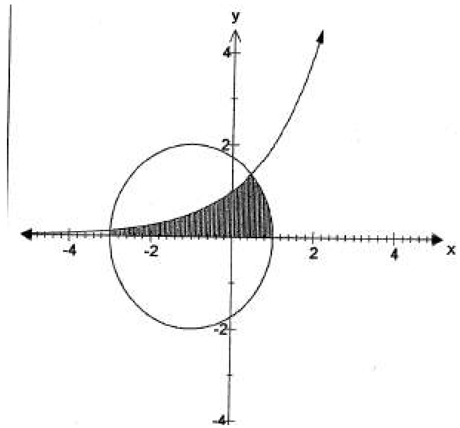


Domain: $-5 \le x \le 5$

Range: $-5 \le y \le 5$

Q22 Show the region of the number plane where the following hold simultaneously: **3**

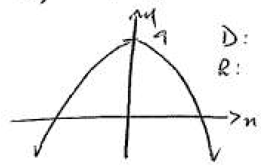
$(x+1)^2 + y^2 \le 4$ $y \le 2^x$ and $y \ge 0$



Q23 On separate number grids sketch the following functions showing all essential features. For each graph state its domain and range. **4**

i). $y = 9 - x^2$ ii) $xy = 4$

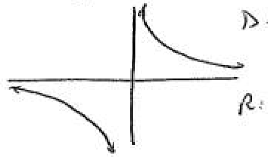
9i) $9 - x^2$



D: x all real

R: $y \le 9$

ii) $xy = 4$

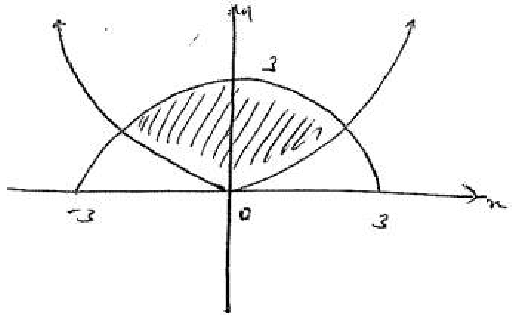


D: x all real except $x \ne 0$

R: y all real except $y \ne 0$

Q24 Draw on the same sketch diagram the curves $y = x^2$ and $y = \sqrt{9 - x^2}$. Indicate on your diagram, by shading, the region which satisfies all the inequalities: **3**

$$y \geq 0;\ \ y \geq x^2 \text{ and } y \leq \sqrt{9 - x^2}.$$

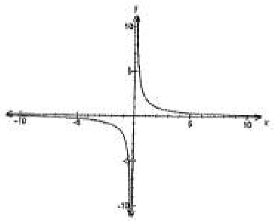


Q25 Sketch the graph for each of the following.
State the domain and range for each graph.

(i) $y = \dfrac{2}{x}$ **3**

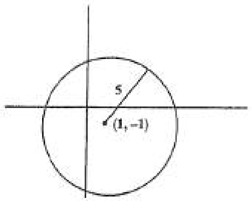
(ii) $(x - 1)^2 + (y + 1)^2 = 25$ **3**

(i)



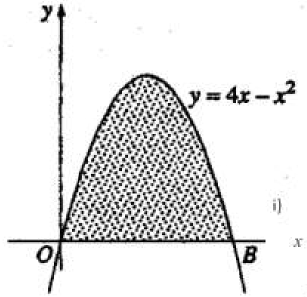
Domain: All real x; $x \neq 0$
Range: All real y; $y \neq 0$

(ii)



Domain: $-4 \leq x \leq 6$
Range: $-6 \leq y \leq 4$

Q26 The diagram shows the graph of the function $y = 4x - x^2$



i) Find the range of the function. **2**

ii) Write down a pair of inequalities that specify the shaded region. **2**

i) Find the range of the function.

$x = \dfrac{-b}{2a}$

$= \dfrac{-(4)}{2(-1)}$

$= 2$

When $x = 2$,

$y = 4(2) - (2)^2 = 4$

Therefore the range is

$y \leq 4$

i) The shaded region will be below the graph and above the line $y = 0$.
Therefore the inequality is $y \leq 4x - x^2$ and $y \geq 0$

Q27 Find the domain of $y = \dfrac{1}{\sqrt{64 - x^2}}$. **2**

$y = \dfrac{1}{\sqrt{64 - x^2}}$

$\sqrt{64 - x^2}$ exists when

$64 - x^2 \geq 0$

$64 \geq x^2$

$x^2 \leq 64$ 2

$-8 \leq x \leq 8$

D: $-8 \leq x \leq 8$

Q28 (a) For each of the following curves

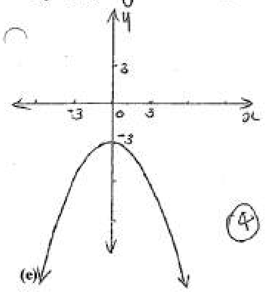
(α) $y = -x^2 - 3$ (β) $y = |x+1|$

(γ) $y = \dfrac{1}{x+2}$ (δ) $y = \sqrt{7+x}$ 16

(i) Draw a neat sketch showing important features.

(ii) State the domain and range.

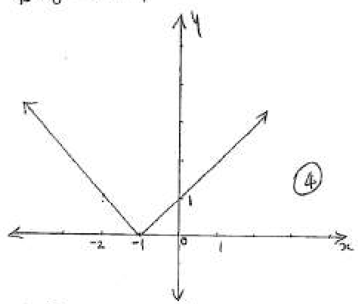
(a) (α) $y = -x^2 - 3$



D : all real x

R : $y \le -3$, y real

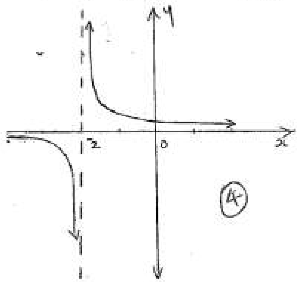
(β) $y = |x+1|$



D : all real x

R : $y \ge 0$, y real

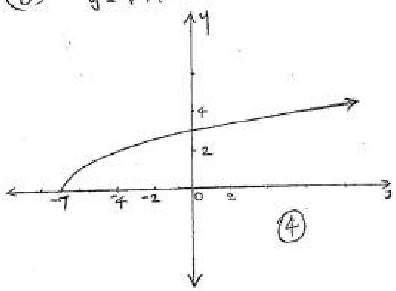
(γ) $y = \frac{1}{x+2}$ page 6



D : all real x, $x \ne -2$

R : all real y, $y \ne 0$

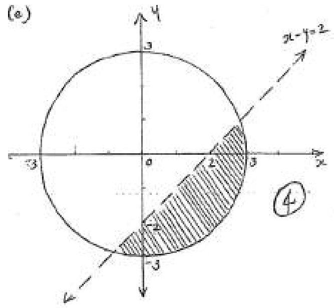
(δ) $y = \sqrt{7+x}$



D : $x \ge -7$

R : $y \ge 0$

Q29 Shade the region given by $x^2 + y^2 \le 9$ and $x - y > 2$. 4



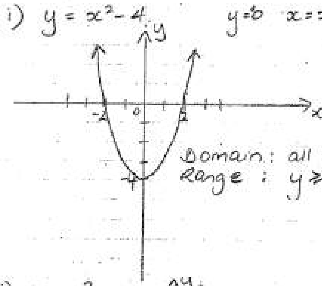
Q30 Sketch the following functions, showing any intercepts with coordinate axes and stating their domain and range.

(i) $y = x^2 - 4$ 4

(ii) $y = \dfrac{2}{x}$ 3

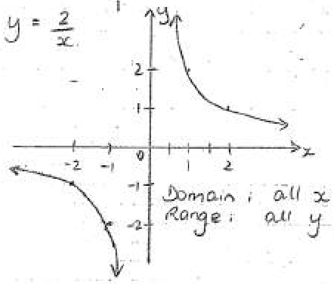
(iii) $y = \sqrt{5-x}$ 4

a) i) $y = x^2 - 4$ $y=0$ $x = \pm 2$.



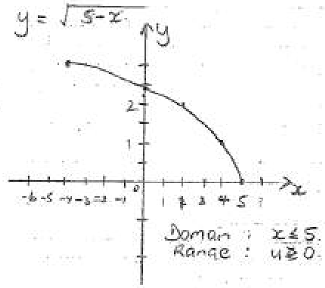
Domain : all real x.
Range : $y \ge -4$.

ii) $y = \frac{2}{x}$



Domain : all x $x \ne 0$.
Range : all y $y \ne 0$.

iii) $y = \sqrt{5-x}$

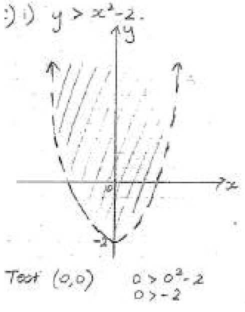


Domain : $x \le 5$.
Range : $y \ge 0$.

Q31

Shade the region corresponding to each of the following inequalities on separate number planes. (i) $y > x^2 - 2$ **3** (ii) $xy \le 1$ **4**

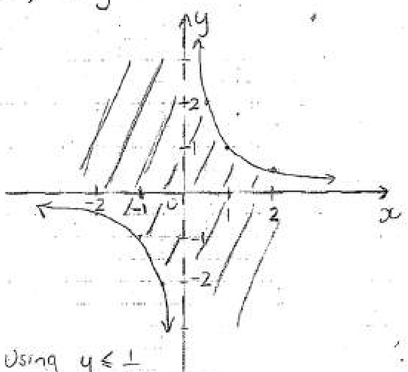
(i) $y > x^2 - 2$.



Test $(0,0)$ $0 > 0^2 - 2$

$0 > -2$

(ii) $xy \le 1$.



Using $y \le \frac{1}{x}$

Test $(0,0)$ $0 \le \frac{1}{0}$ undefined.

Using $xy \le 1$

Test $(2,1)$ $2 \le 1$ not satisfied.

Test $(-2,-2)$ $4 \le 1$ not satisfied.

Q32

Find the domain of $y = \dfrac{1}{\sqrt{25 - x^2}}$. 2

(b) $y = \dfrac{1}{\sqrt{25-x^2}}$

$\sqrt{25-x^2}$ exists when

$25 - x^2 \ge 0$

$25 \ge x^2$

$x^2 \le 25$

ie $-5 \le x \le 5$

D : $-5 < x < 5$

(2)

Q33

For each of the following curves 16

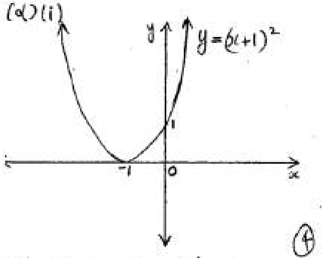
(α) $y = (x+1)^2$ (β) $y = |x| - 3$

(γ) $y = \dfrac{1}{x-4}$ (δ) $y = \sqrt{2-x}$

(i) Draw a neat sketch showing important features.

(ii) State the domain and range.

(α)(i)

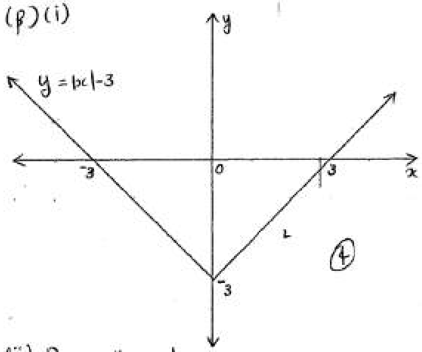


(4)

(ii) D : all real x

R : $y \ge 0$, y real

(β)(i)

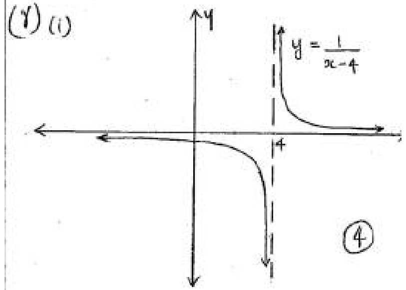


(4)

(ii) D : all real x

R : $y \ge 3$, $y \in R$

(γ)(i)

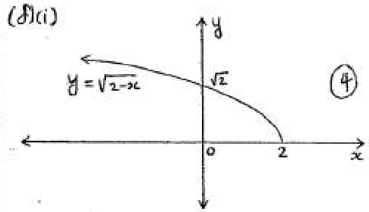


(4)

(ii) D : all real x, $x \ne 4$

R : all real y, $y \ne 0$

(δ)(i)



(4)

(ii) D : $x \le 2$

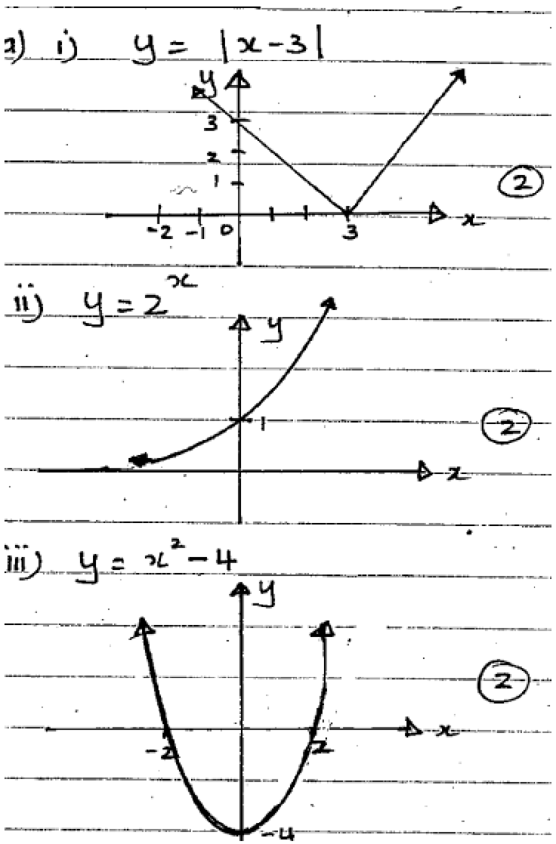
R : $y \ge 0$

Q34 Draw separate sketches of the following, showing the important features:

(i) $y = |x-3|$ **2**

(ii) $y = 2^x$ **2**

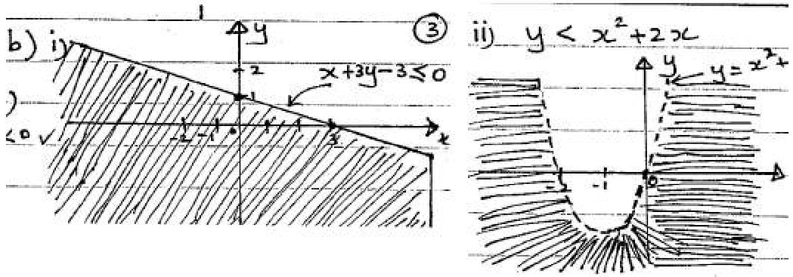
(iii) $y = x^2 - 4$ **2**



Q35 Indicate clearly on a diagram the region determined by the inequality:

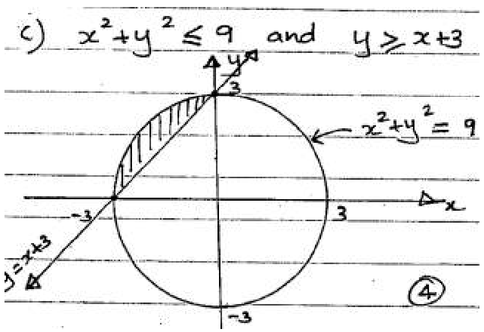
(i) $x + 3y - 3 \le 0$ **3**

(ii) $y < x^2 + 2x$ **3**



Q36 Draw a clear sketch of the region which satisfies the inequalities

$x^2 + y^2 \le 9$ and $y \ge x + 3$ **4**

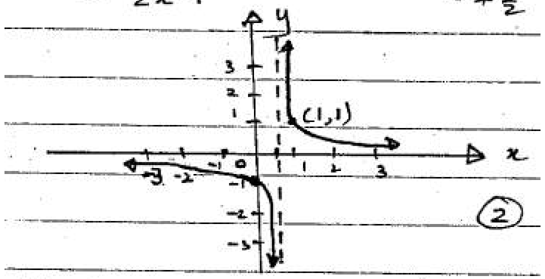


Q37

(i) Sketch the graph of $y=\dfrac{1}{2x-1}$ clearly showing any special features. 2

(ii) State the domain and range of the function. 2

d) i) $y=\dfrac{1}{2x-1}$ $\quad 2x-1\neq 0$, $x\neq \frac{1}{2}$



ii) Domain : $x\in R$, $x\neq \frac{1}{2}$

Range : $y\in R$, $y\neq 0$ 2

Q38

For the function $y=\sqrt{9-x^2}$, (i) Find the domain. 2

(ii) Find the range. 1

(i) Domain $-3\le x\le 3$ 2

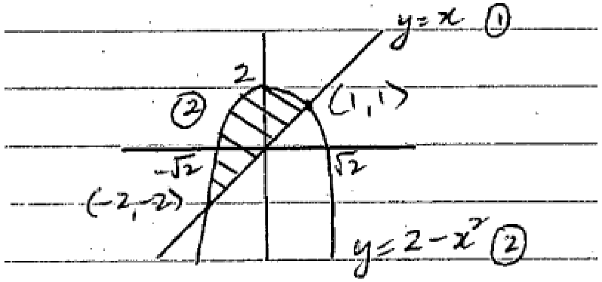
(ii) Range $0\le y\le 3$ 1

Q39

On the same set of axes, sketch the graph of:

(i) $y=2-x^2$. 2 (ii) $y=x$. 1

(iii) Shade the region where $y\le 2-x^2$ and $y\ge x$. 2



Q40

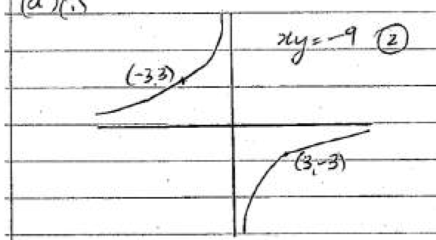
Draw a sketch graph for each of the following functions:

(i) $xy=-9$. 2

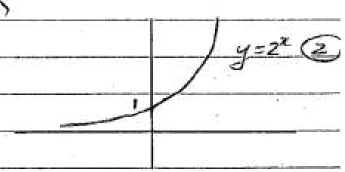
(ii) $y=2^x$. 2

(iii) $x^2+y^2=4$ 2

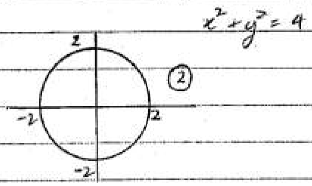
(d)(i)



(ii)



(iii)



Q41

The domain and range for the fu

A. $x \leq 7$, all real y

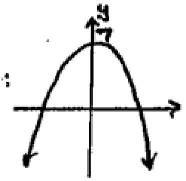
B. $x \leq \sqrt{7}$, all real y

C. All real x, $y \leq 7$

D. All real x, $y \leq \sqrt{7}$

From the graph :
D: all real x
R: $y \leq 7$

C



Q42

Determine the natural domain of $\dfrac{1}{x^2 - 9}$ 2 m

$\dfrac{1}{x^2-9}$ D: $x^2 - 9 \neq 0$

$x^2 \neq 3$

$x \neq \pm 3$

Q43

State the domain and range of $y = x^2 + x - 6$ **2**

(b) Domain is all real x

A.O.S $x = -\frac{1}{2}$ $y = (-\frac{1}{2})^2 - \frac{1}{2} - 6$

$y = -6\frac{1}{4}$

$\therefore$ range is $y \geq -6\frac{1}{4}$

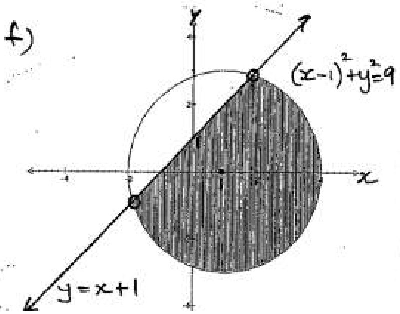
Q44

Show the region of the number plane where the following hold simultaneously:

$(x-1)^2 + y^2 \leq 9$ 3

$y < x + 1$

f)



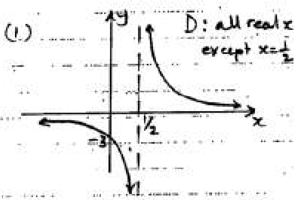
Q45

Sketch graphs of the following functions and state the **domain** of each.

(i) $y = \dfrac{3}{2x-1}$. 2

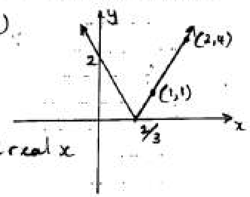
(ii) $y = |2 - 3x|$. 2

e) (i)



D: all real x except $x = \frac{1}{2}$

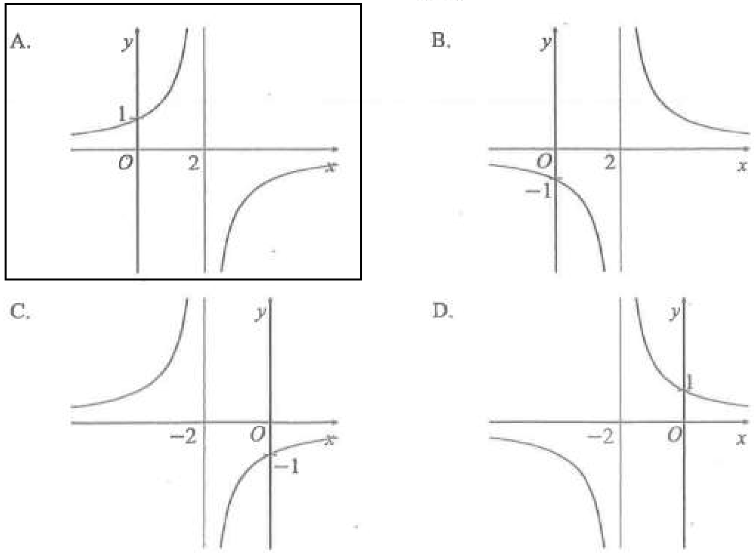
(ii)



D: all real x

Q46

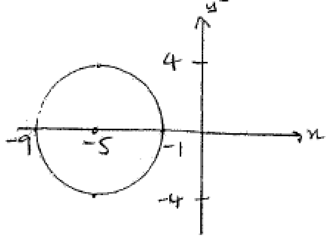
Which of the following graphs represents $y=\dfrac{2}{2-x}$?



Q47

State the domain of: $(x+5)^2+y^2=16$ 1

(c) $(x+5)^2+y^2=16$

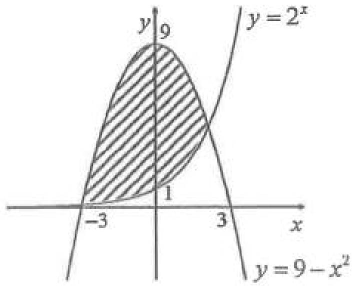


$C(-5,0)$
$r=4$

$D: -9 \le x \le -1$

Q48

Which pair of inequalities defines the shaded region?



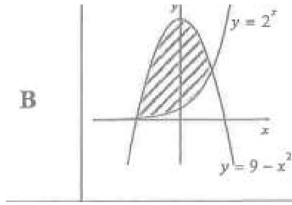
A. $\begin{cases} y \ge 9-x^2 \\ y \ge 2^x \end{cases}$

B. $\begin{cases} y \le 9-x^2 \\ y \ge 2^x \end{cases}$

C. $\begin{cases} y \ge 9-x^2 \\ y \le 2^x \end{cases}$

D. $\begin{cases} y \le 9-x^2 \\ y \le 2^x \end{cases}$

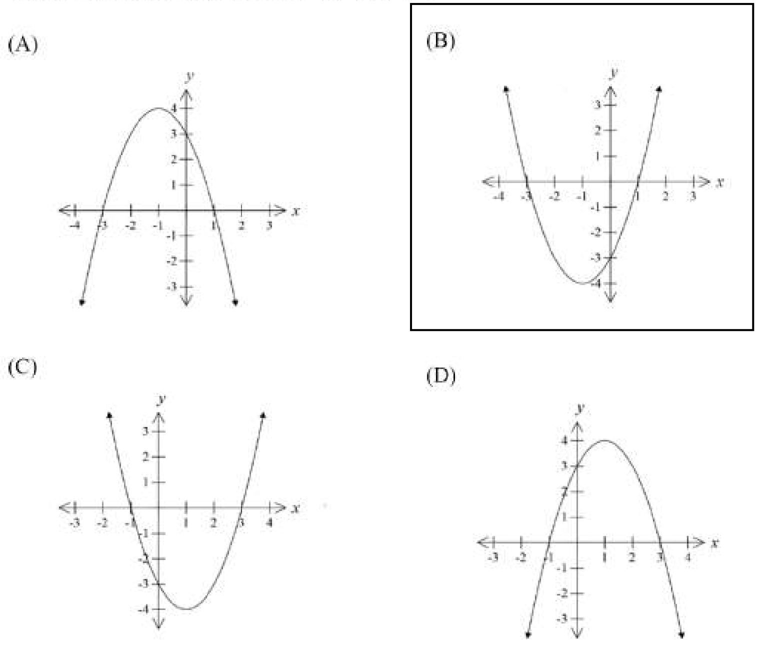
B



$\begin{cases} y \le 9-x^2 \\ y \ge 2^x \end{cases}$

Q49

Which graph best represents $y=x^2+2x-3$?



Q50

What is the domain and range of the function $f(x)=\sqrt{9-x^2}$?

$\boxed{\text{(A) Domain: } -3 \le x \le 3\text{, Range: } 0 \le y \le 3}$

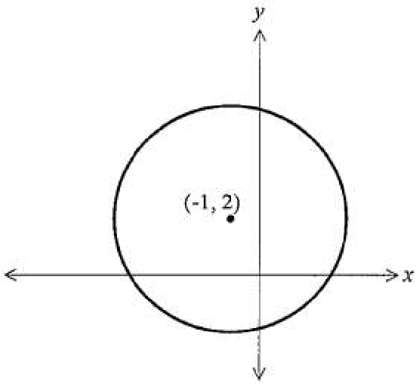
(B) Domain: $-3 \le x \le 3$, Range: $-3 \le y \le 3$

(C) Domain: $0 \le x \le 9$, Range: $-9 \le y \le 9$

(D) Domain: $0 \le x \le 9$, Range: $0 \le y \le 9$

Q51

The graph shows the circle centred at $(-1, 2)$. Choose the correct equation for the circle.



(A) $(x-1)^2 + (y+2)^2 = 1$

(B) $(x+1)^2 + (y-2)^2 = 1$

(C) $(x-1)^2 + (y+2)^2 = 16$

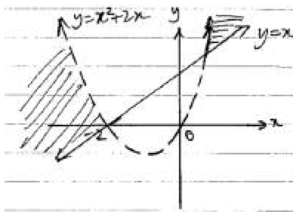
$\boxed{\text{(D) } (x+1)^2 + (y-2)^2 = 16}$

(D) centre $(-1, 2)$ and radius greater than 1.

Q52

Sketch the region that satisfies both $y < x^2 + 2x$ and $y - x \ge 2$. **3**

(c) $y < x^2+2x$ boundary graph $y = x(x+2)$
Test $(0,-1)$ $-1 < 0+0$ True.
$y - x \ge 2$ boundary $y = x+2$ Test $(-3, 0)$
$0-(-3) \ge 2$
$3 \ge 2$ true.



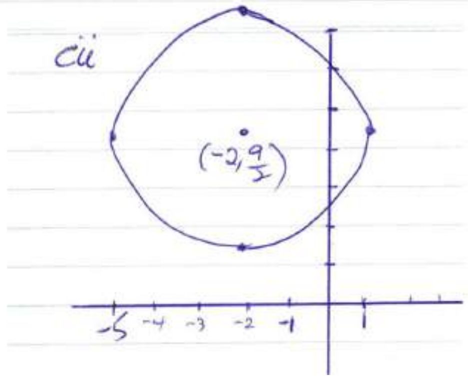
1 for each graph
1 for region.
(3)

Q53

(i) Write $x^2+4x+y^2-9y+4=0$ in the form $(x-h)^2+(y-k)^2=r^2$ **2**

(ii) Hence sketch $x^2+4x+y^2-9y+4=0$ on a number plane. **2**

i). $x^2+4x+y^2-9y+4=0$
$(x^2+4x+4)+(y^2-9y+\frac{81}{4}) = -4+4+\frac{81}{4}$
$(x+2)^2+(y-\frac{9}{2})^2 = (\frac{9}{2})^2$
Centre $(-2, \frac{9}{2})$ Radius $= \frac{9}{2}$ units



Q54

Sketch the graphs of the following, showing the x and y **intercepts**, stating the **domain** and **range** of each.

i) $y = 2^x$ **4**

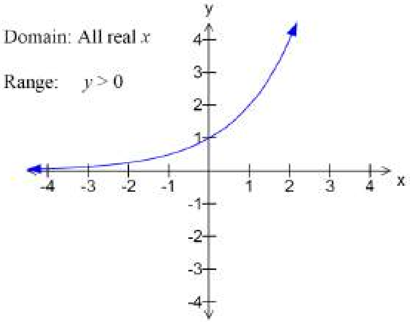
ii) $x^2 + (y+3)^2 = 36$ **4**

iii) $0 = 3x - y - 5$ **4**

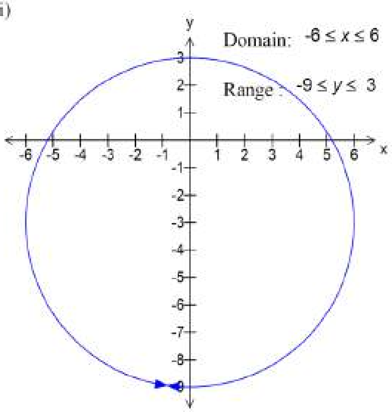
i)

Domain: All real x

Range: $y > 0$



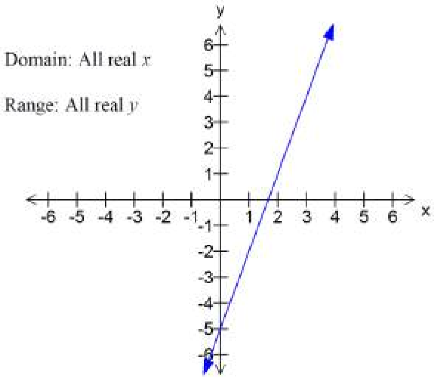
ii)



iii)

Domain: All real x

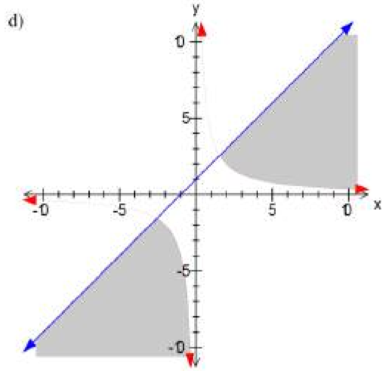
Range: All real y



Q55

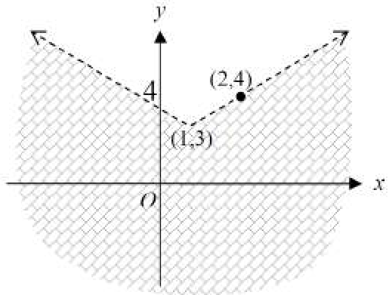
Show the region of the number plane where the following hold simultaneously: $y \le x+1$ and $xy > 4$ **3**

d)



Q56

Write an inequality to describe the shaded region. **3**



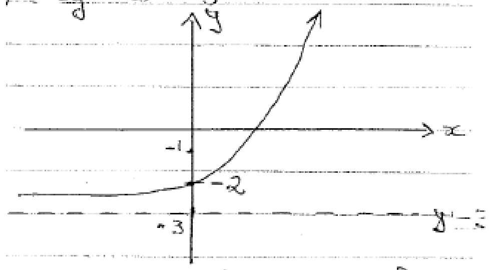
The boundary is $y = |x-1| + 3$

The region is $y < |x-1| + 3$

Or $y < x + 2$ or $y < 4 - x$

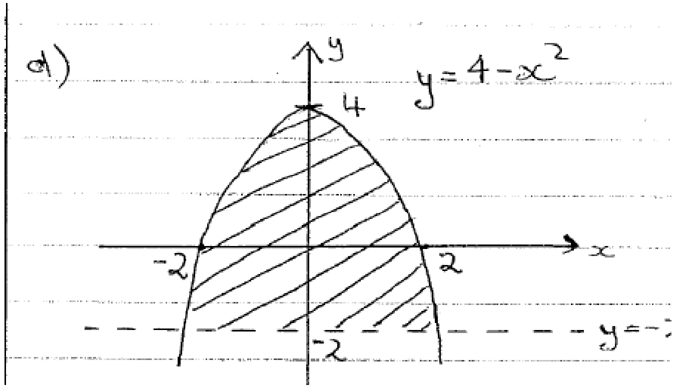
Q57 Sketch $y = 5^x - 3$ and hence state the range of the function. **3**

1) $y = 5^x - 3$

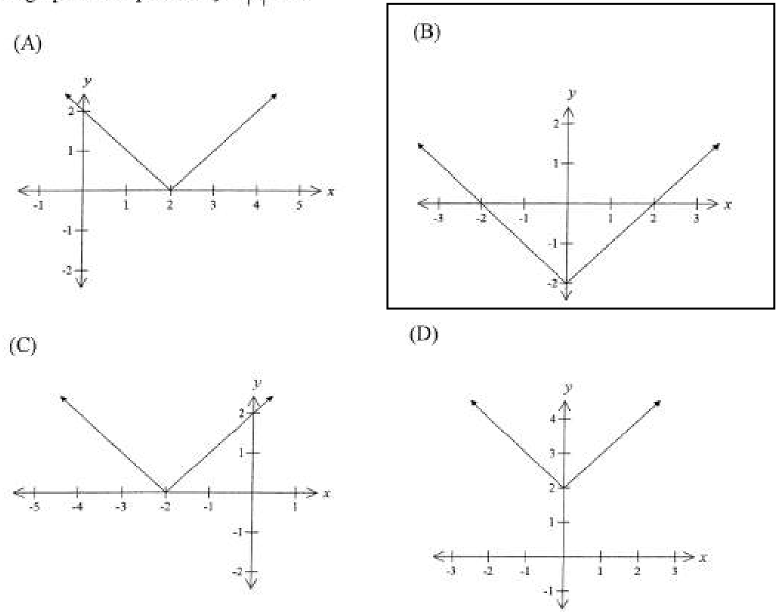


range : $\{y : y > -3\}$

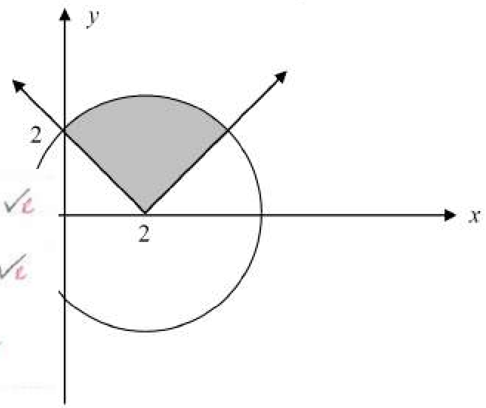
Q58 Sketch the region in the number plane which $y \le 4 - x^2$ and $y > -2$ hold simultaneously. **3**



Q59 Which graph best represents $y = |x| - 2$?



Q60 Write down a pair of inequations to describe the region shaded above. **4**



(b) Absolute value : $y = |x-2|$ ✓

Circle : radius $= \sqrt{2^2+2^2} = \sqrt{8}$ ✓

centre $= (2,0)$

$(x-2)^2 + y^2 = 8$ ✓

Inequations : $y \geq |x-2|$

and $(x-2)^2 + y^2 \leq 8$ ✓

Q 1

If $f(x)=3^x+3^{-x}$

(i) Evaluate $f(0)$ $f(1)$ and $f(-1)$. **2**

(ii) State the domain and range of $f(x)$. **1**

12(a)(i)
$f(0)=3^0+3^0=2$

$f(1)=3^{-1}+3^1=3\frac{1}{3}$, $\quad f(-1)=3^1+3^{-1}=3\frac{1}{3}$

12(a)(ii)
Domain: {x: All real x}
Range: {y: $y\geq 2$}

Q 2

A function $y=f(x)$ is given by the rule

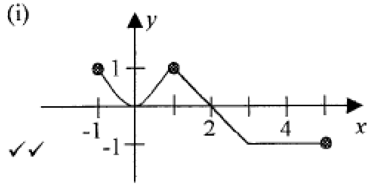
$$f(x)=\begin{cases} x^2 & \text{if } -1\leq x<1 \\ 2-x & \text{if } 1\leq x\leq 3 \\ -1 & \text{if } 3\leq x\leq 5 \end{cases}$$

(i) Graph the function. **2**

(ii) From the graph or otherwise, find the range and domain of the function **2**

(iii) Find the value of $f\left(\frac{1}{2}\right)+f(4)$ **1**

(i)



(ii) $D:-1\leq x\leq 5$ ✓
$R:-1\leq y\leq 1$ ✓

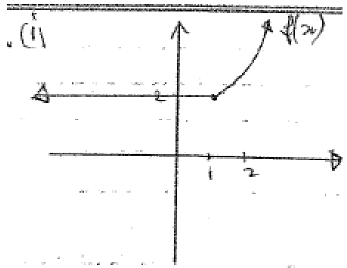
(iii) $f\left(\frac{1}{2}\right)+f(4)=\left(\frac{1}{2}\right)^2-1=-\frac{3}{4}$ ✓

Q 3

Given the function $f(x)=\begin{cases} x^2+1 & \text{for } x\geq 1 \\ 2 & \text{for } x<1 \end{cases}$

i) Sketch the graph of $f(x)$ **2**

ii) Find the value of $f(2)$ **1**



ii) $f(2) = 2^2+1$
$= 5$

Q 4

If $f(x)=\begin{cases} -4 & \text{for } x\leq -2 \\ x & \text{for } -2<x<1 \\ x^2-1 & \text{for } x\geq 1 \end{cases}$

i) Find $f(-1)$ **1**

ii) Find $f(a^2+1)$ **2**
Simplify your answer.

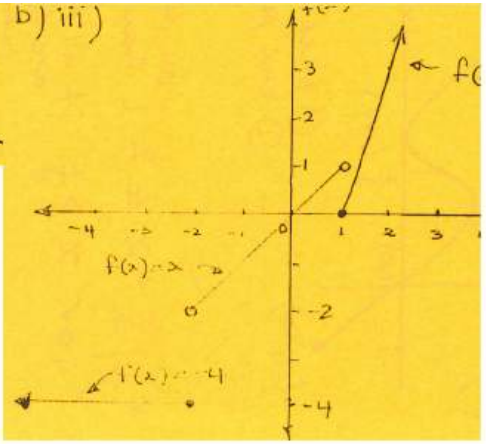
iii) Sketch the graph of $f(x)=y$. **3**

b) i) $f(-1) = -4$ ✓

ii) $f(a^2+1) = (a^2+1)^2-1$ ✓

$= a^4+2a^2+1-1 = a^4+2a^2$

b) iii)



Q 5

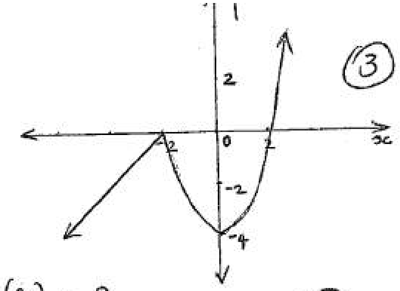
A function is defined by the rule

$$g(x)=\begin{cases} x^2-4 & x\ge -2 \\ x+2 & x<-2 \end{cases}$$

(i) Sketch the curve. 3

(ii) Find the value of $g(2)$, and $g(-3)$. 2

d)



③

$g(2) = 0$

$g(-3) = -3+2 = -1$ ②

Q 6

Show that the following function is odd:

$$f(x)=x^3+4x$$

Hence, or otherwise, evaluate $f(2)+f(-2)$ 3

b) $f(x) = x^3+4x$

$f(-x) = (-x)^3 + 4(-x)$

$= -x^3 - 4x$

$= -(x^3+4x) = -f(x)$

$\therefore f(x) = -f(x)$

$\therefore$ The function is odd.

$f(2) = 2^3+4(2)$

$= 8+8$

$= 16$

$f(-2) = (-2)^3 + 4(-2)$

$= -8-8$

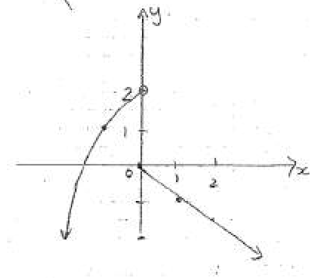
$= -16$

$\therefore f(2)+f(-2) = 16-16 = 0$

Q 7

Sketch $y=\begin{cases} 2-x^2 & x<0 \\ -x & x\ge 0 \end{cases}$ 3

$y = \begin{cases} 2-x^2 & x<0 \\ -x & x\geq 0 \end{cases}$



Q 8

A function is defined by the rule

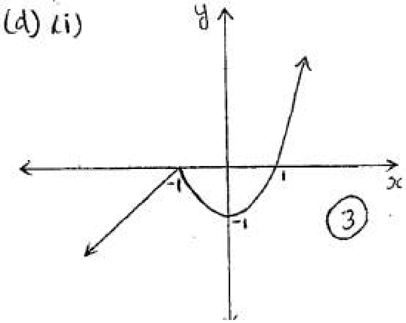
$$f(x)=\begin{cases} x^2-1 & x\ge -1 \\ x+1 & x<-1 \end{cases}$$

(i) Sketch the curve. 3

(ii) State the domain and range. 2

(iii) Find the value of $f(2)+f(-2)$. 2

(d) (i)



③

(ii) Domain : R

Range : R ②

(iii) $f(2)+f(-2)$

$=(4-1)+(-2+1)$

$= 3-1 = 2$ ②

Q 9

A function is defined as: $f(x)=\begin{cases}2x-1 & for\, 0\le x\le 3\\ \frac{1}{3}x+4 & for\, 3<x\le 5\end{cases}$

(i) Find the domain of the function. 1

(ii) Find the range of the function. 1

(iii) Find the value of $f(4)-f(2)$. 2

(a) (i) Domain $0\le x\le 5$ ①

(ii) Range When $x=0$, $y=-1$

When $x=5$, $y=\frac{5}{3}+4=5\frac{2}{3}$

Range $-1\le y\le 5\frac{2}{3}$ ①

(iii) $f(4)-f(2)$

$=(\frac{1}{3}\times 4+4)-(2\times 2-1)$

$=5\frac{1}{3}-3$

$=2\frac{1}{3}$ ②

Q10

Given $f(x)=\sqrt{(x-1)(x-2)}$,

(i) find $f(3)$ 1

(ii) find the domain of $f(x)$ 2

i, $f(3)=\sqrt{2\cdot 1}$

$=\sqrt{2}$ ← ① for correct answer

ii, $(x-1)(x-2)\geq 0$

② 1 for each answer

$\therefore$ Domain : $x\le 1$, $x\geq 2$.

Q11

A function is defined as:

$$f(x)=\begin{cases}x-5 & for\;\; x\ge 0\\ -2 & for\;\; -1<x<0\\ 2+x^2 & for\;\; x\le -1\end{cases}$$

i. Evaluate $f(2)+f(-1)$ 1

ii. On a one third page number plane neatly sketch the function $y=f(x)$ clearly indicating all important features. 2

iii. Write an expression for $f(a^2)$ 1

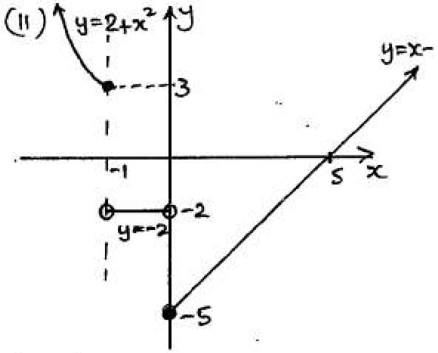
$\therefore$ (i) $f(2)=2-5=-3$

$f(-1)=2+(-1)^2=3$

$\therefore f(2)+f(-1)=-3+3$

$=0$

(ii)



(iii) $f(a^2)=a^2-5$

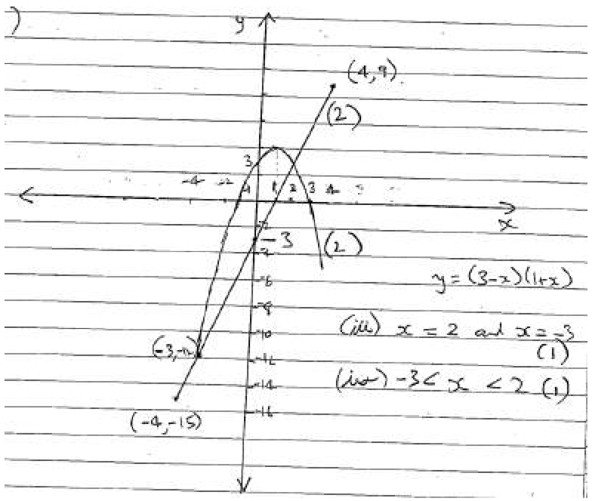
Q12

(i) Sketch the function $y = 3(x-1)$ in the domain $-4 \le x \le 4$, indicating both the x and y intercepts. **2**

(ii) On the same set of axes sketch $y = 3 + 2x - x^2$, showing intercepts **2**

(iii) Deduce the **x-values** of the points of intersection of the two curves **1**

(iv) Hence, or otherwise, solve $3(x-1) < 3 + 2x - x^2$ **1**



(iii) $x = 2$ and $x = -3$ (1)

(iv) $-3 < x < 2$ (1)

Q13

Given $F(x) = \sqrt{2-x}$

(i) Find $F(-3)$ (1)

(ii) Find x if $F(x) = 0$ (1)

(iii) State the domain and range of the function $y = F(x)$ (2)

(iv) Sketch the curve $y = F(x)$ (2)

(v) Find the point of intersection of the curve $y = F(x)$ and the line $x + y = 0$ (3)

e) i) $F(-3) = \sqrt{5}$

ii) $0 = \sqrt{2-x}$

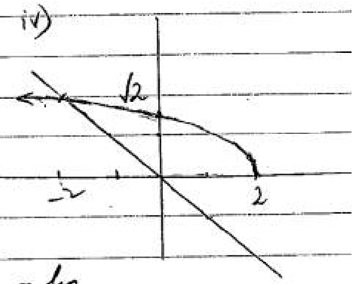
$0 = 2 - x$

$x = 2$

iii) D: $x \le 2$

R: $y \ge 0$

iv)



v) $-x = \sqrt{2-x}$

$x^2 = 2 - x$

$x^2 + x - 2 = 0$

$(x+2)(x-1) = 0$

$x = -2$ or 1

Only $x = -2$ works

Point of intersection is $(-2, 2)$

Q14

A function is defined by the rule:

$$f(x)=\begin{cases} x^2+4 & x\ge 0 \\ \dfrac{1}{x} & x<0 \end{cases}$$

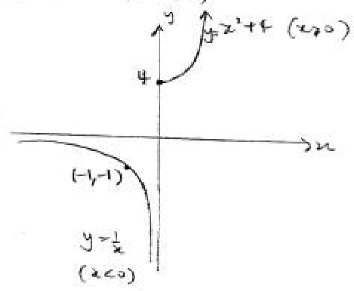
(i) Find $f(k^2)$ 1

(ii) Sketch $y=f(x)$ showing all its features. 2

(d) $f(x) = \begin{cases} x^2+4 & x\geq 0 \\ \frac{1}{x} & x<0 \end{cases}$

(i) $f(k^2) = (k^2)^2+4$
$= k^4+4$

(ii) Sketch $y=f(x)$



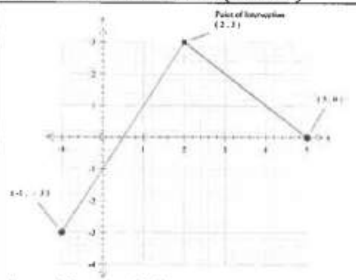
Q15

If $g(x)=\begin{cases} 5-x, & x\ge 2 \\ 2x-1, & x<2 \end{cases}$

(i) Find the value of $g(2)-g(-2)$ 2

(ii) Sketch the graph of $y=g(x)$, $-1\le x\le 5$ 2

(iii) Hence, or otherwise, solve the equation $g(k)=1$ 2

(i)	8	$g(2)-g(-2)=3-(-5)$ *(1 mark)* $=8$ *(1 mark)* CORRECT ANSWER ONLY *(2 marks)*
(ii)	Graph	 *1 mark* for graph shape. *1 mark* for point of intersection and end points MUST BE MARKED.
(iii)	$k=1$ or $k=4$	One mark each value of k.

Q16

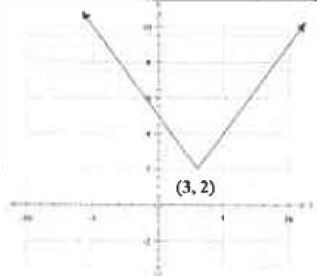
For the function $f(x)=|x-3|+2$

(i) Find $f(-3)$. 1

(ii) State the domain and range of $y=f(x)$. 2

(iii) Explain why $f(x)=0$ has no solution. 1

(iv) Sketch the function $y=f(x)$. 2

(b)	(i)	8	Correct Answer Only, by substitution.
	(ii)	$D: x\in R$ $R: y\geq 2$	*1 mark* each answer.
	(iii)	$f(x)$ does not	intersect with $y=0$, or similar.
	(iv)	Graph	 (3, 2) *1 mark* for absolute value shape. *1 mark* for (3, 2). Point need not be marked.

Q17

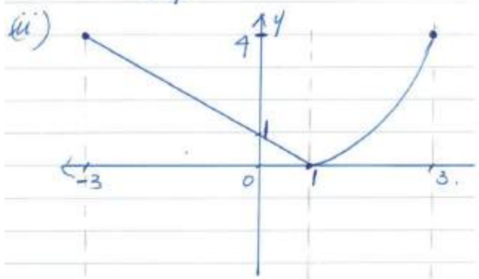
If $f(x)=\begin{cases}(x-1)^2, & x\geq 1\\ 1-x, & x<1\end{cases}$

(i) Find the value of $f(3)$ and $f(-3)$. **2**

(ii) Sketch the graph of $y=f(x)$ in the domain $-3\leq x\leq 3$. **3**

i) $f(3) = (3-1)^2$
$= 4$
$f(-3) = 1-(-3)$
$= 4$

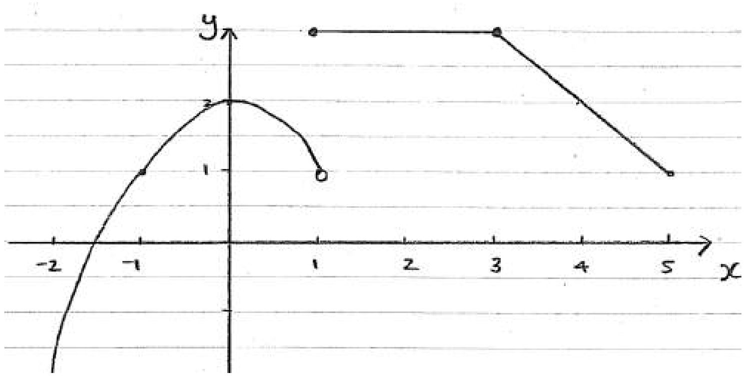
ii)



Q18

Sketch the following piecemeal function in the domain $-2\leq x\leq 5$ **4**

$$f(x)=\begin{cases}-x^2+2 & \text{for } x<1\\ 3 & \text{for } 1\leq x\leq 3\\ 6-x & \text{for } x>3\end{cases}$$



Q19

A function $g(x)$ is the function defined by

$$g(x)=\begin{cases}-1 & \text{for} & x\leq -1\\ x^2-2 & \text{for} & -1<x<1\\ -\dfrac{1}{x} & \text{for} & x\geq 1\end{cases}$$

i) State the range of $g(x)$. **2**

ii) Evaluate $g(2)-2g(0)+g(-3)$ **3**

c) i) $-2\leq y<0$

ii) $g(2)-2g(0)+g(-3)=-\dfrac{1}{2}-2\left(0^2-2\right)-1$

$=2\dfrac{1}{2}$

Q20

For the function defined by:

$$f(x)=\begin{cases}3-x^2 & \text{for} & -2\leq x\leq -1\\ 2x & \text{for} & -1<x<1\\ x^2+1 & \text{for} & 1\leq x\leq 2\end{cases}$$

(i) Evaluate:

(α) $f(-2)$ **1**

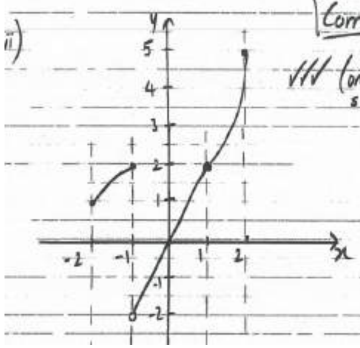
(β) $f(1)$ **1**

(ii) Sketch the graph of the function in the given domain. **3**

i α) $f(-2) = 3-4$
$= 1$

β) $f(1) = 1^2+1$
$= 2$

ii)



$\boxed{\text{Com } 3}$

/// (one per section)